Palak Pandey

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Education

Pillai College of Engineering, University of Mumbai — New Panvel, Maharashtra
B.Tech, Information Technology Engineering (AI/ML focus) • CGPA: 8.6/10 — *Expected 2027*

Sainath Junior College — Navi Mumbai, Maharashtra
HSC (Science, PCMB) • 75.8% — *2023*

Technical Skills

Programming: Python, SQL, Java, C++, JavaScript, TypeScript
Web & App Development: HTML, CSS, React, Tailwind CSS, Flask, Streamlit, Android (Java, XML)
Data Analytics: Excel (Pivot Tables, VLOOKUP, SUMIFS, COUNTIFS), Power BI (DAX, Power Query), Pandas, NumPy, Matplotlib, Data Cleaning, EDA, Dashboard Development
Machine Learning & AI: Scikit-learn, Random Forest, XGBoost, Feature Engineering, Model Evaluation, TensorFlow/Keras, CNN, MobileNetV2 Transfer Learning, TF-IDF, LangChain, RAG, ChromaDB
Cloud & Tools: AWS fundamentals (academic coursework), Git, GitHub

Experience

Data Analyst Intern — Jul 2026 – Aug 2026
HappieLoop Technologies — *Remote*

- Analyzed retail data of 3,900+ customers using Python, SQL, and Excel; performed data cleaning, preprocessing, and exploratory data analysis.
- Built a Power BI dashboard covering KPIs, customer segmentation, sales trends, subscription status, age groups, product categories, retention trends, and average purchasing behavior.

Projects

Social Media Virality Prediction – Multimodal Machine Learning | *Python, Scikit-learn, XGBoost, TensorFlow/Keras, MobileNetV*

- Built a multimodal pipeline combining image, text, and tabular features to predict the virality of social media posts.
- Extracted image features using MobileNetV2 transfer learning and text features using TF-IDF; fused all modalities with an XGBoost classifier.
- Fusion model achieved 86% accuracy and 0.904 ROC-AUC on a 100-sample test set.

Nykaa vs Tira vs Purplle – Beauty & Cosmetics Market Intelligence Dashboard | *Python, SQL, Power BI, Excel, Pandas, NumI*

- Cleaned and prepared 166,665 records using Python, Pandas, SQL, and Power Query for exploratory analysis.
- Analyzed pricing, discounts, brand comparisons, and customer behavior across the three platforms; built an interactive Power BI dashboard using DAX.
- Trained a Random Forest model on campaign data ($R^2 = 0.77$).

Pizza Sales Analysis Dashboard | *SQL, Microsoft Excel*

- Analyzed 50,000+ order records with SQL to compute KPIs: total revenue, total orders, pizzas sold, average order value, and average pizzas per order.
- Built an Excel dashboard using Pivot Tables, VLOOKUP, SUMIFS, COUNTIFS, and IF functions.

Multilingual Indian Government Schemes Chatbot | *Python, Flask, HTML, CSS, JavaScript*

- Developed a Flask-based chatbot with an `/api/chat` endpoint and an HTML/CSS/JavaScript frontend.
- Provides information on welfare schemes by category, including eligibility, benefits, required documents, and application procedure.

College Knowledge Assistant – RAG Document Q&A | *Python, LangChain, ChromaDB, Llama 3.2 3B Instruct, Streamlit*

- Built a retrieval-augmented generation pipeline: PDF loading, text chunking, embeddings stored in ChromaDB, and semantic retrieval.
- Used Llama 3.2 3B Instruct (Hugging Face) to answer questions from retrieved context; built a Streamlit interface for document upload and Q&A.

Smart Outfit Recommender | *Python, Scikit-learn*

- Trained a Random Forest classifier to predict body type from user measurements (94% accuracy) and built a lightweight Python web interface for outfit suggestions.

Indian E-Commerce Customer Behavior & Retention Analysis | *Python, Scikit-learn, XGBoost*

- Analyzed survey data (269 respondents, 71 attributes) using Chi-Square tests, Random Forest feature importance, and PCA (reduced to 29 features).
- Compared Random Forest and XGBoost classifiers; selected XGBoost after hyperparameter tuning.

Let's Shyp – Logistics Booking Flow | *React, TypeScript, Tailwind CSS*

– Built a 5-stage logistics booking form with real-time price recalculation and state persistence across steps.

Additional Projects

– **Olist E-Commerce Churn Analysis** (SQL, Power BI) | **Beauty Product Cross-Sell & Recommendation Analysis** (Python, Apriori, collaborative filtering, NetworkX; synthetic dataset)

– **Shipment Status Dashboard** (React, JavaScript; conditional rendering for tracking) | **Tracify** (native Android tracking app; Java, Android SDK, XML)

Relevant Coursework

Computer Vision • Software Engineering & Project Management • Cloud Computing (AWS Fundamentals) • Machine Learning & AI • Database Management Systems

Certifications

– Microsoft Learn: Power BI Fundamentals • Microsoft Learn: Data Analytics Achievement

– Data Structures & Algorithms in Java • Generative AI (Simplilearn) • Spoken Tutorial, IIT Bombay (C++, Java, PHP, SQL, Python)

Leadership & Activities

Joint Public Relations (Publicity) Head

Computer Society of India (CSI), Pillai College of Engineering

– Led publicity and communication for a 200+ member chapter; coordinated with the core team on event support and speaker outreach.

– Coordinated the launch of the ByteStream magazine and the chapter's Hacktoberfest participation.

– Participated in the Gen AI Showdown hackathon, building and presenting an AI-powered solution under time constraints.